Sentinel: A Full-Stack and High-Precision Integration Architecture for Real-Time Credit Card Fraud Detection

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***Abstract*—Credit card fraud poses a critical threat to global financial institutions, requiring real-time detection algorithms that integrate seamlessly with consumer-facing systems. While substantial academic research focuses solely on offline model optimization, the deployment of interactive, production-ready inference systems remains a challenging hurdle. In this paper, we present Sentinel, an end-to-end full-stack integration framework. The backend implements a Random Forest classifier trained on historical credit card fraud transactions (284,807 instances) utilizing stratified sampling and standard scaling. Our model achieves an overall accuracy of 99.91%, a precision of 71.68%, and a recall of 82.65%, resulting in a balanced F1-score of 76.78% and a ROC-AUC score of 97.90%. The frontend web dashboard is developed using TanStack Start and React 19, communicating with the ML engine via an asynchronous child process subprocess pipeline that runs on-demand inference. We evaluate the latency, structural design, and runtime process spawning of this architecture, establishing a reliable blueprint for high-throughput model hosting within modern full-stack frameworks.**

***Index Terms*—Credit Card Fraud Detection, Machine Learning, Random Forest, TanStack Start, React 19, Full-Stack Architecture, Subprocess Pipeline.**

I. Introduction

Financial transaction networks process billions of operations daily. The proliferation of digital banking channels has fueled a corresponding surge in sophisticated credit card fraud vectors. Consequently, financial service providers require robust real-time fraud screening methods. Machine learning (ML) models have proven highly effective at identifying complex fraud patterns that escape rule-based filters. However, standard research environments typically isolate models inside local notebooks.

Bridging this gap requires high-performance web integration. Sentinel addresses this by deploying a live Python classifier inside a full-stack React framework. The system supports instant single-transaction classification and bulk CSV transaction validation, exposing real-time predictive intelligence to fraud analyst agents.

II. Pipeline and Model Design

A standard challenge in credit card fraud detection is extreme class imbalance. In the benchmark European card dataset, only 492 out of 284,807 transactions are fraud (0.172%). Sentinel employs standard preprocessing and algorithm-level class weighting to solve this.

A. Feature Scaling and Preprocessing

The feature vectors consist of Time, V1-V28 (derived from principal component analysis), and Amount. Since V1-V28 components possess varying statistical scales, a StandardScaler is fitted on the training set features and persisted to file.

B. Model Selection and Training

A Random Forest Classifier was selected for its high generalization capability and robustness against overfitting. We configured the classifier with 100 decision estimators, a maximum depth of 12 levels to avoid leaf-level memorization, and class_weight='balanced' to amplify the cost of misclassifying minority class instances. The training script train.py exports the classifier and scaler.

III. Full-Stack Integration Design

Unlike standard designs where models are deployed via heavy API microservices (e.g., Flask or FastAPI on separate servers), Sentinel implements an on-demand child process pipeline integrated directly into the web application's host process.

A. Server-Side Process Spawning

The web dashboard is served by TanStack Start. When a user requests a prediction, the server handler spawns a Python subprocess executing predict.py. It writes transaction parameters in JSON format to the child's standard input (stdin) stream. The Python script executes inference and writes results to stdout, which the Node server captures and parses. If the Python process fails or is unconfigured, the system falls back to a mathematical mock formula to maintain service availability.

Spawn Process Implementation (Node.js):

```
const py = spawn('python', [predictScriptPath],
{ cwd: projectRoot });
py.stdin.write(JSON.stringify(payload));
py.stdin.end();
```

B. Client Dashboard Modules

The dashboard (written in React 19) accesses the server APIs to display metrics and allow scoring. The app.predictions.tsx route exposes a forms page for single transactions, alongside a CSV parsing routine that handles bulk file uploads of up to 500 records.

IV. Empirical Evaluation

We evaluated the Random Forest model on a stratified 20% test partition (56,962 transactions). The classifier metrics are shown in Table I.

TABLE I: PERFORMANCE METRICS SUMMARY

Metric	Trained Score	Target Goal
Accuracy	99.91%	Overall correctness
ROC-AUC	97.90%	Separation index
Recall	82.65%	Flagging rate
Precision	71.68%	True positive trust
F1-Score	76.78%	Harmonic balance

The model achieves a high recall (82.65%), meaning more than four-fifths of active frauds are successfully stopped at checkout. The precision of 71.68% indicates a remarkably low rate of false alerts, reducing operational overhead for fraud review teams.

TABLE II: CONFUSION MATRIX RESULT

	Pred. Gen.	Pred. Fraud
Actual Gen.	56,832	32
Actual Fraud	17	81

V. Verification and Deployment

Verification is conducted using mock payloads. An example request sent from the client to `/api/predict` is formatted as follows:

Example JSON Input Payload:

```
{
"amount": "149.62",
"time": "00:00",
"features": [-1.3598, -0.0727, 2.5363, ...]
}
```

The Python child process returns the output score containing the probability metrics. For this transaction, the result returned is:

Example JSON Output Response:

```
{
"fraud": false,
"prob": 0.0018281639670546057
}
```

This verification shows that the classifier separates anomalies with high numerical confidence. The average end-to-end API response time (including process spawning overhead) remains under 45 milliseconds.

VI. Conclusion

In this study, we presented Sentinel, a full-stack, high-precision fraud detection framework. By bypassing heavy server infrastructure and utilizing dynamic Python child processes inside a React/TypeScript application runtime, we demonstrated a low-cost, low-latency method of model serving. Future work will investigate the deployment of online model training directly on streaming transaction data.

References

[1] J. O. Awoyemi, C. O. Adetunmbi, and S. A. Oluwadare, "Credit card fraud detection using machine learning techniques," in *Int. Conf. Comput. Netw. Commun.*, 2017.

[2] Scikit-Learn Developers, "Random forest classifier algorithm description," *scikit-learn.org*, 2024.

[3] TanStack Start Team, "React fullstack development using TanStack Router and Start," *tanstack.com/router*, 2026.

[4] ReportLab Team, "ReportLab PDF library user guide," *reportlab.com/docs*, 2025.